WORKSHOP PRINT PACK · PROTOTYPE DRAFT

DE PAPIEREN MACHINE

THE PAPER MACHINE

A simple next-word model, built by hand out of the KB's own newsprint — no screens, no power, no software — leaving two public paper archives behind: what was excluded, and what stayed available but unsaid.

Duration	2 hours 30 (150 min), one break
Participants	Designed for 8–10 · maximum 12 · two tables
Facilitators	One. No assistant required.
Language	English facilitation · Dutch material, glossed in English
Space	KB Atelier, ground floor, window-facing
Leaves behind	TWEE SOORTEN STILTE — two window lines, growing across the run

WHAT IS IN THIS PACK

p.2 — Operator's manual · three operations, one parallel operation
p.3 — Station 1 · CORPUS
p.4 — Station 2 · TOKENISE
p.5 — Station 3 · CLASSIFY (parallel)
p.6 — Station 4 · PREDICT
p.7 — Two kinds of silence · take-home · sources
p.8 — Reading-window masks · facilitator pre-cuts the slots
+ separate file — **transcription sheet**, A4 landscape, printed on 2-part NCR
p.9 — Collection-policy cards
p.10–13 — Vocabulary sheets A / B / C / D
p.14–15 — Carrier cards · EXCLUDED and AVAILABLE
p.16–17 — Window headers · print at A3
p.18–21 — Station signs 1–4
p.22 — Excluded box label
p.23 — Take-home colophon cards
p.24 — Facilitator cue sheet · the only page you hold during the session

YOU ARE THE MACHINE TODAY

Today you build a deliberately simple next-word model — a **bigram model** — and expose the human decisions around it. Not a small chatbot: no memory beyond one word, no meaning, no training objective. The smallest thing that can honestly be called a statistical language model, and small enough that eight people can run it with scissors.

THREE OPERATIONS BUILD THE MODEL

1 · CORPUS → what goes in, and what never does
2 · TOKENISE → text becomes countable pieces
4 · PREDICT → sample the next word by frequency

ONE PARALLEL OPERATION, ON THE SAME MATERIAL

3 · CLASSIFY → what can be found, filtered, governed
— it does not build the model

THE STATIONS RUN 1 → 2 → 3 → 4 IN THE ROOM. STATION 3 SITS APART HERE BECAUSE IT FEEDS A DIFFERENT SYSTEM.

Classification is a real operation — most of what a library does, and much of what dataset owners do. It just builds something else: an index, a filter, a set of permissions. Keeping it visibly separate is the point, because public conversations about AI often run the two together.

THE MATERIAL

Printed facsimiles of Dutch newspaper excerpts, 1618–1879, digitised by the KB and published on Delpher. Publications older than 140 years are public domain: you may cut them, mistranscribe them and reassemble them wrongly. Each carries its date and Delpher identifier.

IF YOU DON'T READ DUTCH

You don't need it. Today's operations are scanning, segmenting, counting and sampling. The classification vocabularies are glossed in English. Sit at either table.

THREE RULES

1. **Nothing goes in the bin.** Rejected material goes in the EXCLUDED box; unused material stays in the tail box. The last half hour is about that difference.
2. **Do not fix your mistakes.** The model is built from your handwriting, errors and all.
3. **Write your rule down before you apply it.** A rule you can write down is a rule somebody else can argue with.

THE DECISION YOU ALREADY MADE

At the door you were handed an excerpt and asked to keep or exclude it, with no context and no rule. That was a **preliminary selection, not part of the model** — neither tray feeds the corpus, and both tables start from the same twenty excerpts whatever you chose.

It returns at the end as evidence of something else: how differently a room decides *before* and *after* a policy has been written down.

WHAT YOU TAKE HOME

One word-pair drawn from the finished machine, and a colophon card. Your transcription becomes the model and is cut up; your two carrier cards stay pegged in the window until the end of January.

WHAT THIS DOES NOT DEMONSTRATE

No attention, no embeddings, no learned weights, no training loop. It cannot show how a modern system represents meaning, and proves nothing about one. What it does show at 1:1 scale are the decisions surrounding *any* model: what was collected, how it was segmented, and who described it.

STATION 1

CORPUS

20 MINUTES · YOUR TABLE · OPERATION 1 OF 3

Both tables have the **same 20 excerpts**. Your model has room for **one per person**.

STEP 1 — WRITE THE RULE FIRST (5 MIN)

Before you look properly, agree one sentence that decides what stays. Not a preference — a rule a stranger could apply without you in the room.

OUR COLLECTION POLICY:

_____

_____

STEP 2 — APPLY IT (10 MIN)

Sort into KEEP (one each) and EXCLUDED (all the rest). Excluded goes in the box, face up, not crumpled. If the rule turns out unworkable, don't change it silently — cross it out above and write the version you actually used underneath. That crossing-out is the most interesting mark on this sheet.

EXCERPTS EXCLUDED AT THIS TABLE: _____ · IN THE ROOM: _____

STEP 3 — COUNT WHAT YOU DID (2 MIN)

QUESTION	YOUR TABLE
How many did you keep because they were *legible*, not because of your rule?	
What kind of excerpt did your rule delete every single time?	
Who here would have chosen differently, and on what?	

STEP 4 — COMPARE (3 MIN)

Read the other table's policy. Same twenty excerpts, two different corpora. Note any excerpt that **both tables threw out**:

The KB now does the opposite — but only relatively recently. Founded in 1798; for most of its life what it collected was unsystematic. Only **since 1974** has it kept one copy of every publication appearing in the Netherlands and worked out what things mean later. A training corpus decides first and keeps second. Neither is natural. Both were somebody's policy, written down.

STATION 2

TOKENISE

30 MINUTES · INDIVIDUAL, THEN TABLE · OPERATION 2 OF 3

Everything the model knows comes from this station. Not from the printed page — from your handwriting. Whatever you get wrong here, the machine will say later.

PART A — BE THE OCR (16 MIN)

Take your excerpt, a **reading window** with the slot already cut, and a **transcription sheet** — forty numbered cells on two-part carbonless paper.

- Hold the window against the column. Read only what appears in the slot, moving **left to right, top to bottom**. No going back. No looking at the whole excerpt.
- **One word per cell.** You decide what counts as a word — a word broken across a line in the original, an abbreviation, a price, a date. That decision is the tokenisation. Nobody will correct it.
- Press firmly: the carbon copy underneath is cut differently and you need both.
- You will not finish. Nobody does. The model uses whatever exists when the timer stops.

AFTER THE TIMER — CHECK ONE ROW AGAINST YOUR PRINTED EXCERPT:

CELLS FILLED	WORDS WRONG	WORDS YOU HAD TO JUDGE

Delpher's search runs on machine OCR of the same material and many errors remain in it — nineteenth-century type, broken columns and damaged paper are hard for software too. Keep your printed excerpt: it is the only clean copy in the room.

PART B — CUT IT INTO PAIRS (10 MIN)

Cut on the printed guides — no measuring, no freehand.

- **Top copy:** cut at the orange A marks → pairs 1+2, 3+4, 5+6 …
- **Carbon copy:** cut at the teal B marks → pairs 2+3, 4+5, 6+7 …
- Cut along every row line too. The **ten single-cell offcuts** → EXCLUDED box.

A full sheet gives **35 usable bigrams**: four pairs from the top copy and three from the carbon, per row, five rows. Each scrap is one bigram. Everything into the tray.

What the cutting rule could not use — two different things.

Four transitions are missing — 8→9, 16→17, 24→25, 32→33. Those words sat next to each other in your reading, but they are on different rows and no scissors will make them one scrap. They are genuinely absent from the model.
Ten single-cell offcuts come off the carbon copy at cells 1 and 8 of each row. Those words *do* appear in pairs elsewhere; the offcuts are redundant. Both go in the excluded box, and they record what the rule could not use — the same artefact that breaks text across columns and pages in every digitised archive.

PART C — LOOK AT WHAT YOU MADE (4 MIN)

QUESTION	ANSWER
Roughly how many pairs does your table hold?	
Which word did you have to make a judgement call about?	
Which of your own errors would you most like to take back?	

STATION 3

CLASSIFY

14 MINUTES · WHOLE ROOM, FOUR GROUPS · THE PARALLEL OPERATION

We are pausing the model to operate a different information system on the same material. Nothing here changes a word the paper machine will say. That is not a defect — it is what this fourteen minutes is for. Classification decides what can be *found*, filtered and governed; it does not decide what comes next.

Leave your tables. The room splits into **four groups of two or three**. Everyone works on the **same single document**, and each group gets one system and may use only its terms.

GROUP A	KB-Code era — the KB's own keyword system, in use **1921–1992**
GROUP B	Brinkman era — Depot keywords, in use **1974 → today**
GROUP C	GOO + Basisclassificatie — in use **1980s–2012**
GROUP D	A 2026 annotation queue — 20 s, paid per item

A–C refer to documented KB systems. Sheets A and B are provisional workshop reductions, to be revised with KB expertise. D is a fictional composite.

ROUND 1 — DESCRIBE IT, IN YOUR GROUP (6 MIN)

Agree one description using only your sheet. If nothing fits, you still have to choose — that is the job.

Group D works differently: one person annotates against the twenty-second sand timer while the rest **watch**. The observers' task is the real one — note what gets skipped, guessed, or forced into whichever field will accept it.

OUR GROUP	OUR DESCRIPTION	WHAT WOULDN'T FIT
A / B / C / D		

In one line: **what would this description make findable, and what would it hide?**

ROUND 2 — READ TO THE ROOM, IN ORDER (6 MIN)

1921, then 1974, then the 1980s, then 2026. Four descriptions of one document. None is wrong. All four then pin to the **OTHER DESCRIPTIONS** card — never the EXCLUDED box, because they were never in the model to begin with.

Three schemes accumulated in one catalogue. KB-Code, Brinkman and GOO overlapped in time and did different jobs; the catalogue still carries all three, because old descriptions stop being applied rather than being rewritten. What you can find depends on which scheme was running when a thing arrived.

Now look back at the sorting board. The contrast becomes visible: not one of these four descriptions is on it. Both systems are real, both matter, and they are not the same system.

STATION 4

PREDICT

30 MINUTES · TABLE, THEN ROOM · OPERATION 3 OF 3

PART A — BUILD IT (8 MIN)

File every scrap by its **first** word. Twelve common words have cells on the board — that is the **head**, and it grows thick. Every other first word goes into the **tail box**, in a rough alphabet bin. Stack loose so thickness shows frequency. Don't tidy.

This is a table of transition counts. **Markov built the first one by hand in 1913** — 20,000 letters of *Eugene Onegin*, counted as vowel–consonant pairs, to prove the letters are not independent. Same operation, with words.

PART B — RUN IT AT YOUR TABLE (8 MIN)

1. **The facilitator announces one start word. Both tables use it.** Write it at the left of the output strip — the two runs have to be comparable.
2. Find that word — **board first, then the tail box**. If it is a tail word: go to its alphabet bin, gather every scrap beginning with that exact word, and draw blindly from those matches only — never from the whole bin.
3. Dot the corner of that scrap immediately — you will not remember later which of two identical scraps you used — then write the second word, replace the scrap, look up the new word, repeat.
4. Stop at 15 words or when a word is in neither. Record where it died, then start again from a different word. **Expect dead ends** — your model is very small.

OUR BEST RUN:

_____

RUNS ATTEMPTED: ____ · LONGEST: ____ WORDS · DIED AT: _____

PART C — THE POOLING EXPERIMENT (12 MIN)

First, predict. Circle what merging both tables will do to the longest run:

LONGER · SHORTER · NO RELIABLE DIFFERENCE → because _____

Then merge, physically. Table B carries its head stacks over and drops each into the matching cell on Table A's board. **Table A** merges the six corresponding tail bins. Two minutes, nobody in anybody's way. Sheets are marked **A** or **B**, so provenance stays visible in the merged stacks.

Then test it: start from the **same word the facilitator gave you at the beginning**, and run the pooled model **three times**.

POOLED LENGTHS: ____ / ____ / ____ · TABLE RUNS WERE: ____ / ____

_____

More data gives more possible transitions per word, but guarantees nothing about any single run. **Whether your prediction held is the finding, either way.**

PART D — READ IT OUT (2 MIN)

Read the longest pooled run aloud; circle a word from a transcription error. And be careful with "lost" from here on: every **undotted** scrap, board or box, stayed available the whole time. It simply wasn't drawn — not the same as the excluded box, which never had a chance at all.

A demonstration from 1948. Building on Markov, Shannon did this by hand — open a book at random, find a word, hunt for it elsewhere, write down what follows: "THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD…"

TWO KINDS OF SILENCE

20 MINUTES · EVERYONE · THEN 8 MINUTES TOGETHER

Your machine stayed silent about nearly everything, for **two different reasons**. They go on two different lines.

LINE 1 · EXCLUDED

Never entered the model. It had no chance.

- every excerpt rejected at Station 1
- 4 missing transitions + 10 offcuts
- the door tray: EXCLUDED WITHOUT A RULE

LINE 2 · AVAILABLE

Entered the model. Never drawn. Present the whole time.

- every undotted scrap, board or box
- everything that stayed available, undrawn

Exclusion is a decision somebody made, and you can name who and when. **Non-selection is a probability** — the material is present, simply not chosen. Systems get attacked for the first and defended with the second, and the two are swapped more often than is honest.

And the other door tray: KEPT WITHOUT A RULE. Hold those up against your table's written policy. Would the rule you wrote at 15:20 have kept them? That gap — between what a room chooses on instinct and what it chooses once a policy exists — is the whole reason we made you write the rule down first.

MOUNT ONE OF EACH

Take one object from the excluded pile, and one **undotted** scrap from anywhere in the model — a head stack on the board or a tail bin, both count. Attach each to its **carrier card** — pink for excluded, teal for available — and answer the question printed on it. Write large: these are read from the street.

PINK — WHICH DECISION REMOVED THIS?

_____

TEAL — WHAT MADE THIS POSSIBLE, BUT UNSELECTED?

_____

Peg both to their lines. They stay in the window until the end of January, beside the cards of everyone who came before you. **Take home instead:** one word-pair drawn from the pooled machine, and the colophon card.

THE LAST QUESTION

You have now performed four operations by hand. Which would you automate, which would you assist, and which would you keep human — and who should decide?

Newspapers 1618–1879 · Delpher / KB, public domain — kb.nl/en/research-find/datasets/delpher-newspapers
KB founded 1798; one copy of every Dutch publication since 1974 — kb.nl/en/over-ons/geschiedenis
KB keyword systems (KB-Code, Brinkman, GOO) — kb.nl/onderzoeken-vinden/bronnen/kb-catalogus/trefwoorden
Nederlandse Basisclassificatie, 4th rev. ed. — kb.nl/kbhtml/docs/bc04.pdf
A. A. Markov, statistical investigation of the text of Eugene Onegin, presented 23 January 1913
C. E. Shannon, A Mathematical Theory of Communication, 1948
P. Otlet & H. La Fontaine, Répertoire Bibliographique Universel, 1895–1934

Vocabulary sheets A and B are provisional workshop reductions, not the thesauri. Sheet D is fiction.

READING WINDOW

FACILITATOR: CUT THIS SLOT BEFORE THE SESSION · 60 × 8 mm · SCALPEL AND STEEL RULE



HOLD AGAINST THE COLUMN · MOVE LEFT TO RIGHT ONLY · NEVER BACK

THIS IS YOUR CONTEXT WINDOW. IT IS 60 MILLIMETRES WIDE.

READING WINDOW

FACILITATOR: CUT THIS SLOT BEFORE THE SESSION · 60 × 8 mm · SCALPEL AND STEEL RULE



HOLD AGAINST THE COLUMN · MOVE LEFT TO RIGHT ONLY · NEVER BACK

THIS IS YOUR CONTEXT WINDOW. IT IS 60 MILLIMETRES WIDE.

COLLECTION POLICY

TABLE ____ · ONE SENTENCE ONLY

PIN THIS TO YOUR CORPUS.

COLLECTION POLICY

TABLE ____ · ONE SENTENCE ONLY

PIN THIS TO YOUR CORPUS.

OTHER DESCRIPTIONS

TABLE ____ · STATION 3

Pin the descriptions that did not become the record here. **Not the excluded box** — these were never in the model.

A PARALLEL SYSTEM. IT DECIDES WHAT CAN BE FOUND.

OTHER DESCRIPTIONS

TABLE ____ · STATION 3

Pin the descriptions that did not become the record here. **Not the excluded box** — these were never in the model.

A PARALLEL SYSTEM. IT DECIDES WHAT CAN BE FOUND.

1921 – 1992 · THE KB-CODE CATALOGUER

SHEET A

You catalogue for the KB between the wars. Your vocabulary is broad, hierarchical and confident. There are not many terms, and every publication on earth is supposed to fit one of them. **Use only the words below.**

Godsdienst *religion*	Aardrijkskunde *geography*
Wijsbegeerte *philosophy*	Geschiedenis *history*
Opvoeding en onderwijs *education*	Oudheidkunde *antiquities*
Staatkunde *statecraft, politics*	Letterkunde *literature*
Rechtsgeleerdheid *law*	Taalkunde *linguistics*
Krijgswezen *military affairs*	Schoone kunsten *fine arts*
Zeewezen en scheepvaart *maritime & shipping*	Muziek *music*
Handel en nijverheid *trade & industry*	Huishoudkunde *domestic economy*
Landbouw en veeteelt *agriculture & livestock*	Koloniën *colonies*
Geneeskunde *medicine*	Zedekunde *ethics*
Natuurkunde *physics*	Statistiek *statistics*
Sterrenkunde *astronomy*	Gemengd nieuws *miscellaneous news*

Notice the word "Koloniën". The Delpher newspaper set explicitly includes colonial newspapers. A vocabulary is not neutral: it tells you what an institution treated as one self-evident category. This one has a single word for a quarter of the world.

A provisional workshop reduction in the register of the KB-Code period — not a reproduction of the official thesaurus, and to be revised with KB expertise before delivery. The KB's real historical keyword systems are documented at kb.nl/onderzoeken-vinden/bronnen/kb-catalogus/trefwoorden.

1974 – TODAY · THE BRINKMAN CATALOGUER

SHEET B

You work in the Depot van Nederlandse Publicaties, which since 1974 has indexed every publication appearing in the Netherlands. Your vocabulary is finer, more modern, more social. **Use only the words below.**

economie *economy*	criminaliteit *crime*
arbeidsmarkt *labour market*	rechtspraak *justice*
armoede *poverty*	bestuur *governance*
migratie *migration*	godsdienst *religion*
volksgezondheid *public health*	onderwijs *education*
epidemieën *epidemics*	vrouwengeschiedenis *women's history*
scheepvaart *shipping*	landbouw *agriculture*
havens *ports*	techniek *technology*
internationale handel *international trade*	weerberichten *weather reports*
kolonialisme *colonialism*	advertenties *advertisements*
slavernij *slavery*	familieberichten *family announcements*
oorlog en conflict *war & conflict*	dagelijks leven *everyday life*

You have words Sheet A does not have. *slavernij*. *migratie*. *vrouwengeschiedenis*. The documents did not change between 1921 and 1974. The vocabulary did. Ask what that does to what a reader can look for — and notice that this comparison is between two reduced lists, not the full historical thesauri.

A provisional workshop reduction in the register of the Brinkman period. Real Brinkman subject terms are published as KB linked data at data.bibliotheken.nl and should be substituted before delivery.

1980S — 2012 · THE CLASSIFICATION SCHEME

SHEET C

You do not use words. You use numbers, from the **Nederlandse Basisclassificatie**. Every document must receive a code. There is no cell for "advertisement", "shipping notice" or "death announcement" — those are not fields of scholarship. **Assign a code anyway.**

01 Algemene werken *general works*	30 Exacte wetenschappen *sciences*
02 Wetenschap en cultuur *science & culture*	31 Wiskunde *mathematics*
05 Communicatiewetenschap *communication*	33 Natuurkunde *physics*
06 Documentaire informatie *documentation*	35 Scheikunde *chemistry*
08 Filosofie *philosophy*	38 Aardwetenschappen *earth sciences*
10 Geesteswetenschappen *humanities*	39 Astronomie *astronomy*
11 Theologie *theology*	42 Biologie *biology*
15 Geschiedenis *history*	43 Milieukunde *environment*
17 Taal- en literatuurwetenschap *linguistics*	44 Geneeskunde *medicine*
18 Taal- en letterkunde *languages*	46 Diergeneeskunde *veterinary*
20 Kunstwetenschappen *art studies*	48 Landbouwwetenschappen *agriculture*
21 Afzonderlijke kunstvormen *art forms*	49 Huishoudkunde *home economics*
24 Theater- en muziekwetenschap *theatre, music*	50 Technische wetenschappen *engineering*

The vocabulary does not fit the document. That is not a fault in your sheet — it is the ordinary condition of every labelled dataset. A scheme built to shelve academic monographs is being asked to describe a shipping advertisement from 1841. Something has to give, and it is never the scheme.

These are the real top-level hoofdgroepen of the Nederlandse Basisclassificatie, 4th revised edition, as published by the KB at kb.nl/kbhtml/docs/bc04.pdf. Print the full list from there if you also want the social-science range 70–89.

2026 · A FICTIONAL COMPOSITE SCENARIO

SHEET D

You are not a cataloguer. You are working through a queue. **20 seconds per document**, timed by you, on the sand timer. You are paid per item, not per hour. You cannot ask anyone a question, and you do not know what the labels will be used for.

FIELD	OPTIONS — CHOOSE EXACTLY ONE
TOPIC	news · commerce · politics · religion · personal · other
SENTIMENT	positive · neutral · negative
SAFETY	safe · sensitive · unsafe
QUALITY	1 2 3 4 5
LANGUAGE	nl · other · unreadable

YOUR ENTRY

TOPIC	SENTIMENT	SAFETY	QUALITY	LANGUAGE

There is no field for "I don't know". None for "this is in a language I can't read", "this needs context I don't have", or "this category is wrong for this object". The form only accepts answers.

Sheets A, B and C describe real systems built by people with job titles, training and time. **This sheet is fiction** — a composite drawn from publicly described annotation practice, written for this workshop. It is not a document from any company, and no claim is made that this work replaced library cataloguing. What it makes you feel is the evidence; the form is not.

EXCLUDED

NEVER ENTERED THE MODEL

ATTACH YOUR OBJECT HERE

WHICH DECISION REMOVED THIS?

WRITE LARGE — THIS IS READ FROM THE STREET · DE PAPIEREN MACHINE · KB ATELIER

EXCLUDED

NEVER ENTERED THE MODEL

ATTACH YOUR OBJECT HERE

WHICH DECISION REMOVED THIS?

WRITE LARGE — THIS IS READ FROM THE STREET · DE PAPIEREN MACHINE · KB ATELIER

AVAILABLE

IN THE MODEL. NEVER DRAWN.

ATTACH YOUR SCRAP HERE

WHAT MADE THIS POSSIBLE,
BUT UNSELECTED?

WRITE LARGE — THIS IS READ FROM THE STREET · DE PAPIEREN MACHINE · KB ATELIER

AVAILABLE

IN THE MODEL. NEVER DRAWN.

ATTACH YOUR SCRAP HERE

WHAT MADE THIS POSSIBLE,
BUT UNSELECTED?

WRITE LARGE — THIS IS READ FROM THE STREET · DE PAPIEREN MACHINE · KB ATELIER

TWEE SOORTEN STILTE · TWO KINDS OF SILENCE · LINE 1 · PRINT AT A3

EXCLUDED

Never entered the model.
Somebody decided.

DE PAPIEREN MACHINE · KB ATELIER
EACH CARD WAS MOUNTED BY A PARTICIPANT
THE LINE GROWS WITH EVERY SESSION

TWEE SOORTEN STILTE · TWO KINDS OF SILENCE · LINE 2 · PRINT AT A3

AVAILABLE BUT UNSAID

Was in the model all along.
Simply not selected.

DE PAPIEREN MACHINE · KB ATELIER
EACH CARD WAS MOUNTED BY A PARTICIPANT
THE LINE GROWS WITH EVERY SESSION

STATION

01

CORPUS

20 excerpts in.
One each out.
Write the rule before you use it.

20 MIN · OPERATION 1 OF 3 · THE REST GOES IN THE EXCLUDED BOX

STATION

02

TOKENISE

Read only through the slot.
Never go back.
Everything the model knows
comes from your handwriting.

30 MIN · OPERATION 2 OF 3 · DO NOT FIX YOUR MISTAKES

STATION

03

CLASSIFY

One document.
Four groups.
A real operation —
on a different system.

14 MIN · WHOLE ROOM · IT DOES NOT BUILD THE MODEL

STATION

04

PREDICT

File every pair by its first word.
Draw blind. Dot it. Replace it.
Then pool the room
and test your prediction.

30 MIN · OPERATION 3 OF 3 · MARKOV DID THIS BY HAND IN 1913

TABLE _____

EXCLUDED BOX

Only material that **never entered the model**, face up, uncrumpled.

Excerpts nobody chose.
Offcuts and missing transitions.
The door tray: kept and excluded without a rule.

Not this box:
· scraps in the tail box that were never drawn — a different kind of silence, and it has its own line
· descriptions from Station 3 — they were never in the model at all, and they pin to the OTHER DESCRIPTIONS card

DE PAPIEREN MACHINE

You built a bigram model from Dutch newspapers printed before 1880 — out of your own handwriting — and you kept what it excluded separate from what it simply never said.

Do it again anywhere: take any book, a pencil, and Shannon's 1948 method — open a page at random, pick a word, find that word elsewhere, write down what follows it, repeat. That is a bigram model. Markov counted the first transition table by hand in 1913.

MATERIAL: DELPHER / KB, PUBLIC DOMAIN
STAPLE YOUR WORD-PAIR HERE ↓

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MATERIAL: DELPHER / KB, PUBLIC DOMAIN
STAPLE YOUR WORD-PAIR HERE ↓

CUE SHEET

150 MIN · ONE FACILITATOR · TWO TABLES · 8–10 PEOPLE, 12 MAX

CLOCK	WHAT HAPPENS	YOU DO / YOU SAY
−45	Set up	Two lines strung + headers hung. Boxes labelled. Slots pre-cut. Excerpts face down. Sheets in the folder.
00 15:00	Door decision · names	"Keep it, or exclude it?" Nothing else. Then: *you just made the first decision, without knowing what for.*
10 15:10	Framing	Index card → Mundaneum (13 m cards, 1,500 queries/yr, warned over 50 results) → "three operations, plus one parallel one." Three rules. "No Dutch needed."
20 15:20	**1 CORPUS**	5′ rule · 10′ sort · 5′ swap. **Watch:** tables abandoning their rule — make them cross it out. Close on 1798/1974.
40 15:40	**2 TOKENISE**	16′ transcribe (call 8/4/1) · 10′ cut on the guides · 4′ check. "One word per cell — *you* decide what a word is."
70 16:10	Break · reset	8′. Lay out the four group packs + the shared document. Swap the station sign.
78 16:18	**3 CLASSIFY**	Four groups of 2–3, one document. 6′ describe · 6′ read 1921→2026. Say "fiction" before D. Close: *the contrast becomes visible* — walk them to the board.
92 16:32	**4 PREDICT**	**Announce one start word for both tables.** 8′ sort · 8′ run + dot every scrap when drawn · 12′ predict → B carries head stacks to A, A merges tail bins → 3 pooled runs from that same word. Do not promise it goes further.
122 17:02	**TWO SILENCES**	20′. Empty excluded boxes + both door trays. Tail boxes stay full. Name the difference, hold up KEPT WITHOUT A RULE against the policies, 12′ carrier cards, peg.
142 17:22	Closing · 8′	"Automate, assist, or keep human — and who decides?" Then hand out take-homes and stop talking.
150 17:30	Strike	20′. Lines stay up. Photograph both, plus the pooled board. Keep the policy cards.

THE THREE THINGS THAT GO WRONG

1. "So the labels are the training data." Correct it warmly and at once. Sequence, not labels.

2. Tail scraps treated as discards. Stop and reset — the closing collapses otherwise.

3. Chains dying at two words. Pool early and start the pooled runs from a populated head word. **Never add outside data** — a short run is a result.

TABLE ROLES — ASSIGN AT 15:20, ROTATE AT THE BREAK

Timekeeper — watches the clock so you don't have to
Material keeper — boxes, tray, nothing goes in the bin
Retriever — fetches and returns sheets from the folder
Recorder — fills the counts on the worksheets

These replace an assistant. Give them out by name, out loud.

If you are running ten minutes late at the break: take it from Station 4 part B — one table runs aloud while the other watches, then pool as planned. **Never shorten the closing.** The two lines are the whole argument and the only thing that outlives the afternoon.